

The empirical recurrence for OEIS A211470, proved by a one-dimensional reduction

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Abstract

OEIS A211470 counts symmetric $(n+1) \times (n+1)$ matrices over $\{-8, \dots, 8\}$ in which every 2×2 subblock sums to zero and has one, two or three distinct entries. Both sides of the matrix grow with n , so there is no slice of bounded width to walk along. There does not need to be: the sum-zero condition is linear and solves completely, and its solutions are indexed by a SEQUENCE — the diagonal of the matrix. The count becomes a walk on $S = 1044$ states, and the empirical recurrence of order 31 contributed by R. H. Hardin follows.

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1 The conjecture

OEIS A211470 is “Number of $(n+1) \times (n+1)$ $-8..8$ symmetric matrices with every 2×2 subblock having sum zero and one, two or three distinct values.”. It has offset 1 and begins

145, 397, 945, 2153, 4775, 10527, 22995, 50565, ...

The entry states, as a conjecture contributed by its author and never marked settled:

Empirical: $a(n) = 6*a(n-1) + 14*a(n-2) - 144*a(n-3) - 12*a(n-4) + 1553*a(n-5) - 1086*a(n-6) - 9957*a(n-7) + 11142*a(n-8) + 42330*a(n-9) - 59054*a(n-10) - 126094*a(n-11) + 201376*a(n-12) + 271448*a(n-13) - 474492*a(n-14) - 430008*a(n-15) + 794866*a(n-16) + 506763*a(n-17) - 953966*a(n-18) - 447303*a(n-19) + 814746*a(n-20) + 296554*a(n-21) - 485322*a(n-22) - 146850*a(n-23) + 194433*a(n-24) + 52926*a(n-25) - 49234*a(n-26) - 13004*a(n-27) + 7010*a(n-28) + 1900*a(n-29) - 420*a(n-30) - 120*a(n-31)$

As of the “Last modified” line on the live entry (Oct 03 2025 16:11:35, revision 8) this is still recorded as empirical, and nothing on the entry records it as proved.

2 The sum-zero condition solved

Let x be an $N \times N$ symmetric matrix with entries in $\{-8, \dots, 8\}$ such that

$$x(i, j) + x(i, j+1) + x(i+1, j) + x(i+1, j+1) = 0$$

for every $0 \leq i, j \leq N-2$.

Lemma 1. *Such matrices are exactly the matrices*

$$x(i, j) = (-1)^{i+j} \frac{u_i + u_j}{2}, \quad u_0, \dots, u_{N-1} \in \{-8, \dots, 8\} \text{ all of the same parity,}$$

and the sequence is recovered from the matrix by $u_i = x(i, i)$, so the correspondence is a bijection.

Proof. Put $y(i, j) = (-1)^{i+j}x(i, j)$. The four entries of a 2×2 subblock alternate in sign under this substitution, so

$$x(i, j) + x(i, j+1) + x(i+1, j) + x(i+1, j+1) = (-1)^{i+j} [y(i, j) - y(i, j+1) - y(i+1, j) + y(i+1, j+1)],$$

and the condition says exactly that the mixed second difference of y vanishes everywhere. Summing that identity over a rectangle gives $y(i, j) = y(i, 0) + y(0, j) - y(0, 0)$ for all i, j , which is $y(i, j) = f(i) + g(j)$ with $f(i) = y(i, 0) - y(0, 0)$ and $g(j) = y(0, j)$; conversely every such y has vanishing mixed difference. Symmetry of x gives $y(i, j) = y(j, i)$, hence $f(i) - g(i) = f(j) - g(j)$ for all i, j : the difference is a constant c . Writing $u_i = 2(f(i) - c/2)$, which is an integer because $y(i, i) = x(i, i)$ is, we get $y(i, j) = (u_i + u_j)/2$ as claimed. Taking $j = i$ gives $u_i = y(i, i) = x(i, i)$.

For the two side conditions: $u_i + u_j$ must be even for all i, j , which says all the u_i have the same parity; and $|x(i, j)| = |u_i + u_j|/2 \leq 8$ for all i, j is equivalent to $|u_i| \leq 8$ for each i , the case $j = i$ giving one direction and the triangle inequality the other. $\square$

So the matrices of size N correspond one for one to the sequences $u_0, \dots, u_{N-1}$ in $\{-8, \dots, 8\}$ with all entries of one parity. What remains is to say, in terms of u , that every 2×2 subblock has one, two or three distinct entries.

Lemma 2. Write $\pi_i = (u_i, u_{i+1})$ for the i -th consecutive pair. The number of distinct entries of the subblock at (i, j) depends only on π_i and π_j : with $\pi_i = (\alpha, \beta)$ and $\pi_j = (\gamma, \delta)$ it is

$$\left| \left\{ \frac{\alpha+\gamma}{2}, -\frac{\alpha+\delta}{2}, -\frac{\beta+\gamma}{2}, \frac{\beta+\delta}{2} \right\} \right|.$$

Proof. By Lemma 1 the four entries are $(-1)^{i+j}$ times those four numbers, in order. Multiplying all four by ± 1 does not change how many are distinct. $\square$

Call two consecutive pairs COMPATIBLE when that number lies in $\{1, 2, 3\}$. The condition on the matrix is that π_i and π_j are compatible for every i and j — including $i = j$, which asks each pair to be compatible with itself. In other words:

Corollary 3. The matrices counted are exactly the sequences u whose set of consecutive pairs is a clique in the compatibility graph on the ≤ 289 pairs.

3 A walk on the sequence

The clique condition is not local: a pair must be compatible with every pair anywhere else in the sequence. Carrying the set of pairs already used would be an exponential state. Carrying what it FORBIDS is not.

Lemma 4. Let A be the set of pairs compatible with every pair used so far. Then a sequence may be continued exactly by pairs lying in A , and appending a pair π replaces A by $A \cap N(\pi)$, where $N(\pi)$ is the set of pairs compatible with π . Hence the pair (last value of u , A) carries everything the future needs.

Proof. The condition is a conjunction of pairwise compatibilities, so the set of pairs still allowed after using a set T is $\bigcap_{\pi \in T} N(\pi)$, which is a function of T and is updated by intersection as T grows. The next pair is (last value, w), so the last value is the only other thing a step needs. $\square$

The sets A that arise are intersections of neighbourhoods, of which there are far fewer than subsets; enumerated from the start $A_0 = \{\pi : \pi \in N(\pi)\}$, the reachable states number $S = 1044$. The two parity classes never mix, so the walk starts by choosing a parity and a first value, and every state may end a sequence:

$$a(n) = \iota^\top M^j \tau, \quad \tau = \mathbf{1}, \quad S = 1044.$$

